Algebra with functions

In this final review session, we cover the algebra of functions: how to compute the composition, subtraction, addition, and the inverse of functions[6].

1. Addition and Subtraction of Functions

For two functions f and g, with domains D_f and D_g:

$$(f+g)(x) = f(x)+g(x), \quad (f-g)(x) = f(x)-g(x)$$

Domain: The domain of $f+g$ or $f-g$ is $D_f \cap D_g$.

2. Composition of Functions

The composition $f \circ g$ is defined by:

$$(f \circ g)(x) = f(g(x))$$

Procedure:

1. Compute $g(x)$.
2. Apply f to the result.

Domain: The domain of $f \circ g$ consists of all x in D_g such that $g(x)$ is in D_f.

3. Inverse Functions

A function f has an inverse f^{-1} if it is **one-to-one** (passes the horizontal line test).

Finding f^{-1}:

1. Write $y = f(x)$.
2. Interchange x and y.
3. Solve for y.
4. Write $f^{-1}(x) = y$.

Properties:

$$f(f^{-1}(x)) = x \quad \text{and} \quad f^{-1}(f(x)) = x$$

Domain and Range: The domain of f^{-1} is the range of f, and the range of f^{-1} is the domain of f.

[6] January 20

Class Activity

Compute the following operations with functions considering that $f(x) = e^x, g(x) = -(x-1)^2+1$

$$h(x) = (f \circ g)(x)$$
$$j(x) = (g \circ f)(x)$$

Compute: $h^{-1}(x)$

Compute: $\frac{(g \circ j)(x)}{2}$

Solutions of previous class activity:

$$f(x) = \frac{3x^2-1}{x^2-1}.$$

Asymptotes: $y = 3; \quad x = \pm 1$

Domain: $x \in (-\infty,-1) \cup (-1,1) \cup (1,\infty)$

Range: $f(x) \in (-\infty,3) \cup (3,\infty)$

y inteceptions: $f(0) = 1$

x inteceptions: $x = \pm \frac{1}{\sqrt{3}}$

$$g(x) = \log\left[2e^{x-4}-2\right].$$

Asymptotes: $x = 4$

Domain: $x \in (4,\infty)$

Range: $f(x) \in (-\infty,\infty)$

y inteceptions: No exist

x inteceptions: $x = \ln\left[\frac{3}{2}+4\right]$

$$h(x) = e^{(x^2-2)/x}.$$

Asymptotes: $y = 0$

Domain: $x \in (-\infty,\infty)$

Range: $f(x) \in (0,\infty)$

y inteceptions: No exist

x inteceptions: No exist